



Original research

Gender-affirming hormone therapy and liver diseases: a cohort study

Newsha Nikzad ,¹ Christina Dimopoulos-Verma,² Seren M Gedallovich,² Douglas Simonetto,³ Alexandra T Strauss,⁴ Casey St Luce,⁵ Sandeep Jain,⁵ C William Pike,⁵ Jananee Muralidharan,⁵ Honor Magon,⁵ Sonali Paul,⁶ Nikki Duong²

► Additional supplemental material is published online only. To view, please visit the journal online (<https://doi.org/10.1136/flgastro-2026-103876>).

¹The University of Chicago Medicine, Chicago, Illinois, USA

²Division of Gastroenterology and Hepatology, Stanford University, Stanford, California, USA

³Division of Gastroenterology and Hepatology, Mayo Clinic, Rochester, New York, USA

⁴Division of Gastroenterology and Hepatology, Johns Hopkins Medicine, Baltimore, Maryland, USA

⁵Atropos Health, New York, New York, USA

⁶Division of Gastroenterology and Hepatology, University of Chicago Medicine, Chicago, Illinois, USA

Correspondence to

Dr Nikki Duong; nduong91@stanford.edu

Received 25 June 2026

Accepted 25 August 2026



© Author(s) (or their employer(s)) 2026. Re-use permitted under CC BY-NC. No commercial re-use. See rights and permissions. Published by BMJ Group.

To cite: Nikzad N, Dimopoulos-Verma C, Gedallovich SM, et al. *Frontline Gastroenterology* Epub ahead of print: [please include Day Month Year]. doi:10.1136/flgastro-2026-103876

ABSTRACT

Objective The purpose of this study was to describe the incidence of liver-related outcomes, including liver disease and hepatic injury, in transgender and gender diverse (TGD) individuals receiving oestrogen-based or testosterone-based gender-affirming hormone therapy (GAHT).

Design/methods In this retrospective cohort study, we used a national electronic health record and claims database (Atropos Health) to describe the incidence of liver-related outcomes in TGD individuals receiving GAHT from 2015 to 2024. Adults with International Classification of Diseases (both versions 9 and 10) diagnosis codes for gender identity disorder/gender dysphoria who had received either oestrogen-based or testosterone-based therapies were included. Those with prior liver disease, alcohol use disorder or endocrine disorders were excluded. Liver enzymes, incident liver disease and metabolic outcomes were assessed at 1 year following initiation of GAHT.

Results Of the 16 379 individuals with available follow-up liver chemistries, 84.9% receiving oestrogen-based GAHT and 82.3% receiving testosterone-based GAHT had normal aspartate aminotransferase and alanine aminotransferase levels at 1 year. In both groups, approximately 2% of patients developed a new diagnosis of steatotic liver disease. New diagnoses of Budd-Chiari syndrome, cirrhosis or hepatocellular carcinoma occurred in <1% of all individuals. Metabolic and thrombotic complications, including venous thromboembolism, hyperlipidaemia and hypertension, were infrequent (1%–3.4%) in both cohorts.

Conclusion This is a large real-world study assessing GAHT and liver-related outcomes. GAHT was associated with low rates of liver enzyme abnormalities, new liver disease and metabolic complications after 1 year.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Gender-affirming hormone therapy is medically necessary for some transgender and gender diverse individuals. The hepatic safety profile of these oestrogen-based or testosterone-based formulations is poorly documented.

WHAT THIS STUDY ADDS

⇒ This large, real-world cohort investigates liver-related outcomes of gender-affirming hormone therapy. Gender-affirming hormone therapy does not appear to be associated with hepatotoxicity or liver diseases after 1 year of therapy.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ These findings support the continued use of gender-affirming hormone therapy (GAHT) as medically necessary care for transgender and gender diverse (TGD) individuals. Ensuring equitable access to GAHT, improving data accuracy and expanding high-quality research will be essential to advancing liver health and reducing disparities for TGD populations.

INTRODUCTION

In the USA, approximately 1.6 million people identify as transgender and an additional 1.2 million people identify as non-binary.^{1 2} Transgender and gender diverse (TGD) individuals in the USA are disproportionately affected by conditions associated with chronic liver disease, such as obesity, diabetes, alcohol use disorder and viral hepatitis.^{3 4} Gender-affirming hormone therapy (GAHT)—comprising feminising regimens (oestrogen with or without androgen blockade) and masculinising regimens (testosterone)—is medically necessary for many individuals experiencing gender dysphoria.⁵

Despite increased utilisation of GAHT, data describing the association between GAHT and liver disease are limited. Existing literature suggests that GAHT is rarely associated with acute or chronic injury in both native and transplanted livers.^{6–12} However, prior studies have been small, heterogeneous, limited to select populations such as transplant patients, or confounded by other causes of liver injury. Thus far, the best currently available evidence demonstrates that oestrogen-based and testosterone-based GAHT have not been associated with liver diseases or injury.¹² Of note, this excludes the oral testosterone formulation named C-17 α alkylated testosterone or methyltestosterone, which is contraindicated in GAHT due to previous studies demonstrating its hepatotoxicity.¹² As GAHT use increases, an improved understanding of its association with liver chemistries and liver-related outcomes is essential for informed counselling, risk stratification and anticipatory guidance in hepatology and transgender health. We conducted a real-world study evaluating liver-related outcomes after initiation of oestrogen-based or testosterone-based GAHT in a national adult TGD cohort.

METHODS

Data source

In this retrospective study, we used a national electronic health record (EHR) and claims database (Apollo, accessed through Atropos Health) to describe the incidence of liver-related outcomes in adults receiving GAHT from 1 January 2015 to 27 December 2024. Given the Apollo database contains entirely deidentified participant data prior to investigator access and did not involve interaction with human subjects, the study was determined to be exempt from review by an Institutional Review Board at Stanford University on 2 February 2025. Participants were not required to give informed consent as the study utilised clinical data without personal identifiers.

Selection criteria

Adult TGD individuals (ages 18–90 years) with International Classification of Diseases (ICD) diagnosis codes for gender identity disorder who had received either oestrogen-based or testosterone-based therapies were included (online supplemental file 1). Specifically, ICD10 codes F64 for descriptor ‘gender identity disorders’ were utilised, including F64.0 ‘transsexualism’, F64.1 ‘gender identity disorder in adolescence and adulthood’, F64.2 ‘gender identity disorder of childhood’, F64.8 ‘other gender identity disorders’ and F64.9 ‘gender identity disorder, unspecified’. ICD9 code 302.85 for descriptor ‘gender identity disorder in adolescents or adults’ was also utilised.^{13 14} Individuals were excluded if they had a prior history of elevated transaminases or liver conditions such as cirrhosis, viral hepatitis B or C, steatohepatitis, Budd-Chiari syndrome, hepatic adenoma or hepatocellular

carcinoma (HCC). Those with any history of alcohol use disorder and/or prior history of HIV, osteoporosis, endocrine disorders or any abnormal hormone levels were also excluded to minimise confounding variables that would make it difficult to attribute any observed liver disease or enzyme changes specifically to GAHT (online supplemental file 1). Demographic data—including age, sex, race and ethnicity—were extracted from EHR. Participant data on baseline body mass index (BMI), lipid panels and any pre-existing diagnoses of hyperlipidaemia, hypertension and venous thromboembolism were also collected.

Data abstraction

The index date was defined as the first GAHT prescription following a diagnosis of gender dysphoria. Individuals were required to have received at least four prescriptions of hormone therapy during the first year from their index date. They were also required to have received GAHT for at least 90 days duration prior to their index date and a follow-up of at least 1-year duration after their index date. The baseline period was defined in two ways, depending on the variable. It was characterised as any time prior to index for binary outcomes, and the 90-day period prior to index date for continuous outcomes. For any patients with two or more continuous outcomes in the 90-day baseline period, the latter of the two was taken.

The nomenclature for steatotic liver disease transitioned from non-alcoholic fatty liver disease/non-alcoholic steatohepatitis (NAFLD/NASH) to metabolic dysfunction-associated steatotic liver disease/metabolic-associated steatohepatitis (MASLD/MASH) in June 2023.¹⁵ However, ICD codes for MASLD and MASH did not yet exist at the time of this study. As a result, ICD-9 and -10 codes for fatty liver, other chronic non-alcoholic liver disease, other specified inflammatory liver diseases and NASH were used to capture patients with steatotic liver disease.

The normal limits for liver chemistries were derived from guideline recommendations: aspartate aminotransferase (AST) <10–32 IU/L and alanine aminotransferase (ALT) <19–25 IU/L in females and AST <10–34 IU/L and ALT <29–33 in males. With regards to elevated liver chemistries, the ranges used for each category were also based on guideline recommendations: borderline elevation is ALT and/or AST greater than 2 $\times$ the upper limit of normal (ULN), mild is 2–5 $\times$ ULN, moderate is 5–15 $\times$ ULN, severe is >15 $\times$ ULN and massive is >10 000 IU/L.¹⁶ Normal ranges were applied as the presumed affirmed sex’s reference range (ie, patients receiving testosterone-based GAHT were evaluated using normal limits for liver enzymes in males).

Data analysis

Outcomes were reported as descriptive statistics in the form of counts and percentage for binary outcomes

(eg, presence or absence of comorbidities) and mean and SD for continuous outcomes (eg, BMI, number of days). Primary outcomes after 1 year of GAHT included AST/ALT levels, new diagnoses of liver disease (NAFLD, NASH, cirrhosis, hepatic adenoma, HCC and Budd-Chiari), or new venous thromboembolism. Secondary outcomes included new diagnoses of hyperlipidaemia, hypertension or new use of prescription statins, antihypertensives or diuretics after 1 year of GAHT. The aforementioned diagnoses were based on ICD-9 and –10 codes. Newly prescribed medications were based on standardised RxNorm codes from the National Library of Medicine.¹⁷ In addition, the BMI and lipid data of participants were described after 6 months of GAHT.

RESULTS

Oestrogen-based GAHT

A total of 7625 individuals initiating oestrogen-based GAHT met inclusion criteria. The mean age was 30.5 years (SD 12.1), with most individuals identifying as white (67.8%), and non-Hispanic or non-Latinx ethnicity (96.6%). Mean baseline BMI was 27.0 kg/m² (SD 7.1). Baseline lipid profiles were within expected ranges for a general adult population (table 1).

Among the 2023 individuals (26.5% of cohort) with recorded AST/ALT measurements at 1 year, 84.9% had normal liver enzyme levels. Borderline or mild-to-moderate elevations occurred in 292 individuals, while severe elevations were not observed. There was a 26.5% (22 023/7625 individuals) versus 24.6% (2150/8754 individuals) laboratory follow-up rates among the oestrogen and testosterone cohorts, respectively. Both cohorts had low overall Charlson comorbidity scores (0.44, SD 1.09 in oestrogen cohort vs 0.31, SD 0.69 in testosterone cohort). The oestrogen cohort was older (mean age 30.5 years, SD 12.1) compared with the testosterone cohort (mean age 26.0 years, SD 8.1). The oestrogen group had a higher baseline hypertension rate of 9.0% and hyperlipidaemia rate of 11.7% in comparison to the testosterone group, which had 5.5% of patients with baseline hypertension and 9.0% with HLD. However, the testosterone cohort had a higher recorded BMI (29.3) compared with that of the oestrogen cohort (27.0). Further comparison of descriptive outcomes among the oestrogen and testosterone groups, respectively, included: abnormal AST/ALT among monitored patients of 15.2% vs 17.8%; incident NAFLD: 1.73% vs 2.07%; incident venous thromboembolism: 1.11% vs 0.63%; and new hyperlipidaemia diagnoses: 3.33% vs 3.37%. Follow-up duration of approximately 1190 days and coded loss to follow-up were nearly identical.

Among the entirety of both cohorts, including patients without AST/ALT measurements, incident steatotic liver disease was uncommon. Rates of Budd-Chiari syndrome (0.01%), venous thromboembolism (1.1%), cirrhosis (0.05%) and HCC (0.01%) were

Table 1 Cohort characteristics based on exposure to therapeutic oestrogen or testosterone

Baseline demographics	Oestrogen N=7625	Testosterone N=8754
Age, mean (SD), years	30.5 (12.1)	26 (8.1)
Gender,* n (%)		
Female	2525 (33.1)	6158 (70.3)
Male	5100 (66.9)	2596 (29.7)
Race, n (%)		
White	5173 (67.8)	6040 (69)
Other†	2002 (26.3)	2105 (24)
Black	275 (3.6)	418 (4.8)
Asian	175 (2.3)	191 (2.2)
Ethnicity, n (%)		
Hispanic or Latinx	259 (3.4)	379 (4.3)
Not Hispanic or Latinx	7366 (96.6)	8375 (95.7)
Start of GAHT, n (%)		
2015–2019	2017 (26.5)	2259 (25.8)
2020–2024	5608 (73.5)	6495 (74.2)
Follow-up days, mean (SD)	1196 (692.9)	1184 (667)
Lost to follow-up, n (%)	7 (0.1)	9 (0.1)
Pre-existing conditions, n (%)		
Hyperlipidaemia	893 (11.7)	784 (9.0)
Hypertension	688 (9.0)	478 (5.5)
Venous thromboembolism	38 (0.5)	34 (0.4)
BMI, mean (SD)	27 (7.1)	29.3 (8.3)
Baseline labs, mean (SD)		
HDL (SD), mg/dL (normal range 40–75)	49.9 (14)	53.6 (14.5)
LDL (SD), mg/dL (normal range 50–130)	100.2 (30.7)	101.4 (30.4)
Triglycerides (SD), mg/dL (normal range 60–150)	153.3 (80)	142.6 (67.6)

*Gender data are extrapolated for electronic health record. Whether it refers to self-reported gender data or assigned sex at birth is unknown and likely dependent on the individual patient and their respective care provider.

†Category label recorded as 'Other' in electronic health record.

BMI, body mass index; GAHT, gender-affirming hormone therapy; HDL, high-density lipoprotein; LDL, low-density lipoprotein.

exceedingly rare. No individuals were newly diagnosed with hepatic adenoma, acute liver injury or drug-induced liver injury. The mean BMI remained stable at 1 year. New diagnoses of hyperlipidaemia (3.3%) or hypertension (1.9%) were infrequent. Approximately 6.7% of individuals were newly started on medications approved for glycaemic control, and 40.8% of individuals were newly started on diuretic medications (tables 2 and 3).

Testosterone-based GAHT

A total of 8754 individuals initiated testosterone-based GAHT. The mean age was 26 years (SD 8.1), with a predominantly white (69%) and non-Hispanic or non-Latinx ethnicity (95.7%) cohort. Mean baseline BMI was 29.3 kg/m² (SD 8.3), and mean baseline

Table 2 Cohort liver-based outcomes after 1 year of exposure to oestrogen or testosterone

Outcomes 1 year after index GAHT	Oestrogen N=7625	Testosterone N=8754
Borderline or mild AST/ALT elevations,* n (%)	292 (14.4)*	374 (17.4)*
Moderate AST/ALT elevations,* n (%)	15 (0.7)	7 (0.3)
Severe AST/ALT elevations, n (%)	0 (0)	1 (0.01)
NAFLD, n (%)	132 (1.7)	181 (2.1)
NASH, n (%)	14 (0.2)	10 (0.11)
Cirrhosis, n (%)	4 (0.05)	5 (0.06)
Hepatic adenoma, n (%)	0 (0)	2 (0.02)
HCC, n (%)	1 (0.01)	1 (0.01)
Budd-Chiari, n (%)	1 (0.01)	2 (0.02)
VTE, n (%)	85 (1.1)	55 (0.6)
New diagnosis HLD, n (%)	254 (3.3)	295 (3.4)
New diagnosis HTN, n (%)	146 (1.9)	178 (2.0)
New prescription statin, n (%)	110 (1.4)	99 (1.1)
New prescription anti-hypertensive, [†] n (%)	407 (5.3)	535 (6.1)
New prescription diuretic, n (%)	3112 (40.8)	79 (0.9)

*Only about ~2000 individuals (N) had recorded ALT/AST.
[†]Includes angiotensin-converting enzyme inhibitors (ACEis), angiotensin II receptor blockers (ARBs), beta blockers and calcium channel blockers (CCBs).
 ALT, alanine aminotransferase; AST, aspartate aminotransferase; GAHT, gender-affirming hormone therapy; GAHT, gender affirming hormone therapy; HCC, hepatocellular carcinoma; HLD, hyperlipidemia; HTN, hypertension; N, number; NAFLD, nonalcoholic fatty liver disease; NASH, nonalcoholic steatohepatitis; VTE, venous thromboembolism.

high-density lipoprotein, low-density lipoprotein and triglyceride levels were within normal ranges (table 1).

Of the 2150 individuals (24.6% of cohort) with recorded AST/ALT measurements, 82.3% had normal liver enzymes. Borderline or mild-to-moderate elevations occurred in 374 individuals, with one individual experiencing severe elevation. Incident hepatic steatosis (2.1%) and steatohepatitis (0.1%) were uncommon. New diagnoses of cirrhosis (0.06%), hepatic adenoma (0.02%), Budd-Chiari syndrome (0.02%) and HCC (0.01%) occurred very infrequently. Metabolic complications were infrequent: hyperlipidaemia (3.4%), hypertension (2.0%) and new glycaemic medication

Table 3 Cohort body mass index and lipid data after 6 months of exposure to oestrogen or testosterone

Outcomes 6 months after index GAHT	Oestrogen N=8672	Testosterone N=8754
BMI, mean (SD)	27.4 (7.3)	29.7 (8.2)
HDL, mean (SD)	54.1 (14.1)	47.0 (12.8)
LDL, mean (SD)	100.5 (31.6)	107.9 (32.5)
Triglycerides, mean (SD)	158 (80.2)	153.3 (77.6)

BMI, body mass index; GAHT, gender affirming hormone therapy; HDL, high-density lipoprotein; LDL, low-density lipoprotein.

use (5.6%). Mean BMI remained stable (29.5 vs 29.3 kg/m²) (tables 2 and 3).

DISCUSSION

This is one of the largest real-world studies to date assessing GAHT and liver safety, including 16 379 individuals using either oestrogen-based or testosterone-based GAHT over 9 years. These findings reinforce and expand on prior large multicentre and other smaller studies demonstrating that GAHT has a generally favourable hepatic safety profile.^{6–12} Hashemi *et al*, the largest US-based longitudinal study (n=1062 transgender individuals across three health systems), found that masculinising GAHT was associated with modest increases in ALT and AST, while feminising GAHT showed no significant liver enzyme changes.⁶ Among the European cohort studies, Stangl *et al* found the incidence of liver injury within 12 months of GAHT initiation was very low (0.1% in transgender women, 0.4%–0.6% in transgender men).¹⁸ Similarly, Boekhout-Berends *et al* found that liver enzymes decreased in transgender women and increased in transgender men after 1 year of GAHT.⁷

Our data showed negligible changes in BMI and low incidences of new hypertension or hyperlipidaemia diagnoses in both groups. Although limited data exist on the cardiovascular and metabolic effects of GAHT, in one large cohort study, testosterone therapy was associated with an increased risk of metabolic syndrome in transmasculine individuals.¹⁹ Other prior studies have not demonstrated an association or causal relationship between GAHT and new diagnoses of hypertension.^{20–21} Some studies have described changes such as alterations in lipid profiles (eg, increased triglycerides) and increased BMI among individuals receiving GAHT, especially among individuals assigned female at birth. However, current data do not demonstrate a significant risk in cardiovascular events.^{22–24}

The observed discordance between low rates of diagnosed hypertension and higher rates of antihypertensive or diuretic prescriptions likely reflects medication use for non-hypertensive indications. While medication analyses were not available to distinguish among diuretics, this finding likely indicates the use of spironolactone for androgen blockade in transfeminine patients rather than treatment of hypertension. However, with the testosterone group, it is unclear why 6.1% of patients received a new prescription of antihypertensives and 0.9% of patients received a new prescription of diuretics when only 2.1% of this group had a new diagnosis of hypertension. This highlights limitations inherent to claims-based analyses and the need for more granular clinical data.

Historically, oestrogen exposure has been associated with venous thromboembolism, Budd-Chiari syndrome and hepatic adenomas, largely in the context of supra-physiologic oestrogen doses such as oral contraceptives in cisgender women.^{5,25–27} In contrast, GAHT regimens

are titrated to achieve physiologic levels of oestrogen and testosterone, and individuals receiving therapy are not exclusively cisgender women.⁵ These differences may explain why our study revealed exceedingly low incidences of venous thromboembolism, Budd-Chiari syndrome and hepatic adenomas in both cohorts.

Limitations

There are several limitations within this study. Approximately a quarter of patients in each cohort had liver enzyme measurements during follow-up, which introduces possible selection bias. There was a 26.5% versus 24.6% laboratory-follow-up rate among the oestrogen and testosterone cohorts, respectively. Our study reflects real-world utilisation patterns of laboratory data, which suggests that there may be low concern among providers for hepatotoxicity secondary to GAHT and therefore, lack of routine surveillance of liver-related outcomes in this patient population. Furthermore, participants with liver enzyme measurements may have experienced clinical signs or symptoms concerning for new pathologies with hepatic involvement, which may have led to disproportionately greater representation of liver enzyme elevation that may not be representative of the overall cohort. Reliance on ICD-9 and ICD-10-based definitions may introduce potential misclassification, particularly for chronic liver disease, although efforts were made to assess fibrosis risk with the available data. However, only 10 patients had the available data needed to calculate a FIB-4 score. Future studies using in-depth chart review with inclusion of labs and imaging may allow for a more accurate assessment of the interaction between GAHT and liver-related diagnoses.

The data in our study came from a predominantly white, non-Hispanic patient population, limiting generalisability of our conclusions about the relationship between GAHT and liver health of a more diverse patient population. In addition, the duration of 1-year follow-up should allow for adequate time to detect new hepatotoxicity and steatotic liver disease from hormone therapy. However, outcomes such as fibrosis progression, cirrhosis, HCC, liver transplantation and cardiometabolic effects of GAHT would likely require a longer follow-up duration than that of this study. Subclinical steatosis, for example, may be underdiagnosed as well.^{28 29} It is also important to note that with inherent limitations in EMR or documentation practices that are not harmonised, the absence of diagnostic codes should not be interpreted as the absence of disease. This suggests the inherent limitation in using diagnostic codes to identify TGD patients and subsequently describe any new diagnoses following GAHT initiation.

Finally, it is important to acknowledge that accurately capturing gender identity in the EHR remains challenging. For example, if the standard was to report sex assigned at birth for this cohort specifically, one

would expect 100% of patients who are receiving oestrogen to be male and 100% of patients who are receiving testosterone to be female. However, there were about 30% of patients in both the oestrogen and testosterone groups that were not characterised in this way (eg, females on oestrogen and males on testosterone). Given this, a possible explanation is that some patients opted to change their sex in the EHR to reflect their lived experience as a TGD individual. Variable coding practices among clinicians, EHR limitations in gender and sex descriptors, and lack of data harmonisation across institutions may have also contributed to this finding. Another possible explanation is that there were some cisgender patients who were receiving hormone replacement therapy for other indications that were included, despite efforts to exclude these patients based on ICD codes.

Future research

Further research is needed to address persistent gaps in our knowledge of GAHT and liver-related outcomes across diverse populations. Comprehensive measures of hepatic safety should be integrated over longer periods of follow-up (beyond 1 year), as many of these individuals may require life-long GAHT. Studies focused on more diverse patient populations are an imperative next step, as significant racial and ethnic diversity exists among TGD individuals. Prior research has shown that Latinx, American Indian or Alaska Native, and biracial and multiracial individuals are more likely than white individuals to identify as TGD.² A dearth of data exists in describing the outcomes for patients receiving GAHT who have undergone a liver transplant, and there is a lack of clear guidelines for GAHT management in the setting of liver transplantation.³⁰

In addition, future studies should include gallstone disease as an outcome and patients with known liver disease, as this patient population is disproportionately affected by liver disease and its risk factors.¹² Prospective studies with specific attention to GAHT formulations, hormone concentrations, cumulative or duration of exposure, standardised laboratory measurements, imaging and fibrosis assessment will be critical to improve phenotyping and establish causal relationships between GAHT and liver outcomes. Efforts should also prioritise improving the accuracy of gender identity documentation in EHRs, addressing recent changes in ICD coding and understanding how policy restrictions on GAHT may influence health-care engagement, research participation and safety for TGD individuals.^{12 30}

CONCLUSION

This large, real-world cohort demonstrates that GAHT is associated with low rates of liver enzyme abnormalities, incident liver disease and metabolic complications during the first year of therapy. These

findings support continued use of GAHT as medically necessary care for TGD individuals. Ensuring equitable access to GAHT, improving data accuracy and expanding high-quality research will be essential to advancing liver health and reducing disparities for TGD populations.

Contributors NN was involved in manuscript concept and design, data acquisition and interpretation and draft of manuscript. CD-V was involved in manuscript concept and design, data acquisition and interpretation and draft of manuscript. SMG was involved in manuscript concept and design, data acquisition and interpretation, and draft of manuscript. DS was involved in data interpretation and critical manuscript revisions. ATS was involved in data interpretation and critical manuscript revisions. CSL was involved in data acquisition and interpretation and accessed and verified data reported in the manuscript. SJ was involved in data acquisition and interpretation and accessed and verified data reported in the manuscript. CWP was involved in data acquisition and interpretation. JM was involved in data acquisition and interpretation. HM was involved in data acquisition and interpretation. SP was involved in manuscript concept and design, data acquisition and interpretation, critical manuscript revisions, and supervision. ND was involved in manuscript concept and design, data acquisition and interpretation, critical manuscript revisions, and supervision. All authors confirm that they had full access to all the data in the study and accept responsibility to submit for publication. ND is the guarantor of the manuscript.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient consent for publication Not applicable.

Ethics approval The Apollo database contained entirely de-identified participant data prior to investigator access and did not involve interaction with human subjects; therefore the study was determined to be exempt from review by an Institutional Review Board (IRB) at Stanford University on 2 February 2025.

Provenance and peer review Not commissioned; externally peer-reviewed.

Data availability statement All data relevant to the study are included in the article or uploaded as supplementary information. Data, analytic methods, and study materials have been made indefinitely available to other researchers through the manuscript's text and tables, which will be made electronically made available upon publication.

Supplemental material This content has been supplied by the author(s). It has not been vetted by BMJ Publishing Group Limited (BMJ) and may not have been peer-reviewed. Any opinions or recommendations discussed are solely those of the author(s) and are not endorsed by BMJ. BMJ disclaims all liability and responsibility arising from any reliance placed on the content. Where the content includes any translated material, BMJ does not warrant the accuracy and reliability of the translations (including but not limited to local regulations, clinical guidelines, terminology, drug names and drug dosages), and is not responsible for any error and/or omissions arising from translation and adaptation or otherwise.

Open access This is an open access article distributed in accordance with the Creative Commons Attribution Non Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited, appropriate credit is given, any changes made indicated, and the use is non-commercial. See: <https://creativecommons.org/licenses/by-nc/4.0/>.

ORCID iD

Newsha Nikzad <https://orcid.org/0000-0001-8447-8527>

REFERENCES

- 1 Wilson BDM, Meyer IH. Nonbinary LGBTQ adults in the United States. The Williams Institute; 2021.
- 2 Herman JL, Flores AR, O'Neill KK. How many adults and youth identify as transgender in the United States? The Williams Institute; 2022.
- 3 Safer JD, Coleman E, Feldman J, *et al*. Barriers to healthcare for transgender individuals. *Curr Opin Endocrinol Diabetes Obes* 2016;23:168–71.
- 4 Chiu I, Cano D, Leathers M, *et al*. HIV and hepatitis C virus infection and co-infection among trans women in San Francisco, 2020. *PLoS One* 2024;19:e0307990.
- 5 Hembree WC, Cohen-Kettenis PT, Gooren L, *et al*. Endocrine Treatment of Gender-Dysphoric/Gender-Incongruent Persons: An Endocrine Society Clinical Practice Guideline. *J Clin Endocrinol Metab* 2017;102:3869–903.
- 6 Hashemi L, Zhang Q, Getahun D, *et al*. Longitudinal Changes in Liver Enzyme Levels Among Transgender People Receiving Gender Affirming Hormone Therapy. *J Sex Med* 2021;18:1662–75.
- 7 Boekhout-Berends ET, Wiepjes CM, Nota NM, *et al*. Changes in laboratory results in transgender individuals on hormone therapy - a retrospective study and practical approach. *Eur J Endocrinol* 2023;lvad052.
- 8 Xu L, Yuan Y, Che Z, *et al*. The Hepatoprotective and Hepatotoxic Roles of Sex and Sex-Related Hormones. *Front Immunol* 2022;13:939631.
- 9 Martin KA, Bostwick JM, Vargas HE. Liver Transplant Case Report: Transgenderism and Liver Transplantation. *Int J Transgend Health* 2011;13:45–9.
- 10 Ramadan OI, Naji A, Levine MH, *et al*. Kidney transplantation and donation in the transgender population: A single-institution case series. *Am J Transplant* 2020;20:2899–904.
- 11 Lee K, Chuzi S, Katz J, *et al*. Heart-Kidney Transplantation in a Transgender Woman. *JACC Case Rep* 2022;4:101523.
- 12 Nguyen TN, Jackson WE, Roth NC, *et al*. Chronic liver disease and hepatology care in transgender and gender diverse populations. *Lancet Gastroenterol Hepatol* 2026;11:334–44.
- 13 American Medical Association. ICD-9-CM, 1996 : international classification of diseases, 9th revision, clinical modification : volumes 1 and 2, color-coded, illustrated ICD-9 CM codes. Dover, DE American Medical Association; 1996.
- 14 World Health Organization. *International statistical classification of diseases and related health problems*. 10th Revision. 2nd edn. Geneva: World Health Organization, 2004.
- 15 Rinella ME, Lazarus JV, Ratzliff V, *et al*. A multisociety Delphi consensus statement on new fatty liver disease nomenclature. *Hepatology* 2023;78:1966–86.
- 16 Kwo PY, Cohen SM, Lim JK. ACG Clinical Guideline: Evaluation of Abnormal Liver Chemistries. *Am J Gastroenterol* 2017;112:18–35.
- 17 RxNorm. National Library of Medicine (US). n.d. Available: <https://www.nlm.nih.gov/research/umls/rxnorm>
- 18 Stangl TA, Wiepjes CM, Defreyne J, *et al*. Is there a need for liver enzyme monitoring in people using gender-affirming hormone therapy? *Eur J Endocrinol* 2021;184:513–20.
- 19 Hashemi L, Marijic Buljubasic A, Budoff MJ, *et al*. Gender-Affirming Hormone Treatment and Metabolic Syndrome Among Transgender Veterans. *JAMA Netw Open* 2024;7:e2419696.
- 20 Martinez-Martin FJ, Kuzior A, Hernandez-Lazaro A, *et al*. Incidence of hypertension in young transgender people after a 5-year follow-up: association with gender-affirming hormonal therapy. *Hypertens Res* 2023;46:219–25.
- 21 Leon SJ, Ferguson TW, Whitlock R, *et al*. Gender-affirming Hormone Therapy and Changes in Kidney Function in

- Adults: A Retrospective Population-based Study. *Kidney Med* 2025;7:101051.
- 22 Rahman SU, Manasrah N, Kumar N, *et al.* Impact of Gender-Affirming Hormonal Therapy on Cardiovascular Risk Factors in Transgender Health: An Updated Meta-Analysis. *JACC Adv* 2024;3:101265.
 - 23 Gois Í, Rodrigues FB, Pereira M, *et al.* Body mass index and body composition changes in transgender people undergoing gender-affirming hormone therapy: a systematic review and meta-analysis. *Rev Endocr Metab Disord* 2025;26:937–53.
 - 24 Moon H, Park S, Lee Y, *et al.* Gender-affirming hormone therapy and cardiovascular risk: an umbrella review of meta-analyses. *Int J Transgend Health* 2026;1–17.
 - 25 García-Pagán JC, Valla D-C. Primary Budd-Chiari Syndrome. *N Engl J Med* 2023;388:1307–16.
 - 26 Wang M, Sarkar M, Mehta N. Now you see it, now you don't: Estrogen exposure and obesity effects on HCA development and regression. *Hepatology* 2023;77:341–3.
 - 27 Sitruk-Ware R. Hormonal contraception and thrombosis. *Fertil Steril* 2016;106:1289–94.
 - 28 Vuppalanchi R, Chalasani N. Drug-induced Liver Injury from Hormonal and Non-hormonal Therapies: Insights from a Large Case Series. *J Clin Exp Hepatol* 2024;14:101401.
 - 29 Clinical course and diagnosis of drug induced liver disease. In: *LiverTox: clinical and research information on drug-induced liver injury*. Bethesda (MD): National Institute of Diabetes and Digestive and Kidney Diseases, 2012.
 - 30 Nikzad N, Fisher AR, Pillai A, *et al.* Considerations in gender-affirming hormone therapy in transgender and gender diverse patients undergoing liver transplantation. *Am J Transplant* 2024;24:1567–72.